Quantum Mechanics I — Exercise Sheet 2

A driven qubit

Variant: Questions only

Conventions, as in Lecture 2: the circular drive $H(t) = \frac{\hbar\nu_0}{2}\sigma_z + \frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y\big)$ with $\nu_0, \Omega, \nu > 0$; $R(t) = e^{-\mathrm{i}\nu t\sigma_z/2}$, $\psi_t = R(t)\tilde{\psi}_t$; $\Delta = \nu_0 - \nu$; in the frame the Hamiltonian is $\tilde{H} = \frac{\hbar}{2}(\Delta\sigma_z + \Omega\sigma_x)$, and the frame Bloch vector $\tilde{\mathbf{r}}$ turns about $(\Omega, 0, \Delta)/\tilde{\Omega}$ at the rate $\tilde{\Omega} = \sqrt{\Omega^2 + \Delta^2}$.

Exercise 1 (another starting point). On resonance, a spin starts in the state $|{+y}\rangle$, with Bloch vector $\mathbf{e}_y$. Find $\tilde{\mathbf{r}}(t)$ and the probability of reading $|{\downarrow}\rangle$ at the time t. When is it first 1? What happens if the spin starts in $|{+x}\rangle$ instead?

Exercise 2 (a π-pulse inverts every state). Show that the π-pulse $-\mathrm{i}\sigma_x$ sends every frame Bloch vector $(\tilde{r}_x, \tilde{r}_y, \tilde{r}_z)$ to $(\tilde{r}_x, -\tilde{r}_y, -\tilde{r}_z)$. What does it do to the probabilities of $|{\uparrow}\rangle$ and $|{\downarrow}\rangle$?

Exercise 3 (why the rotating frame is needed). For a constant H the solution is $e^{-\mathrm{i}tH/\hbar}\psi_0$. A natural guess for the drive is $\psi_T = \exp\big(-\frac{\mathrm{i}}{\hbar}\int_0^T H(t)\,\mathrm{d}t\big)\psi_0$. (i) Compute $\int_0^T H(t)\,\mathrm{d}t$. (ii) On resonance, $\nu = \nu_0$, what does the guess predict for the probability of $|{\downarrow}\rangle$ at $T = 2\pi/\nu$, from $|{\uparrow}\rangle$? What is the true value? Evaluate both for $\Omega/\nu = 0.1$.

Exercise 4 (Ramsey's fringes, shifted). In Ramsey's sequence of Proposition 2.4, the first pulse is driven as in (2.1). The oscillator keeps running through the wait, and for the second pulse its phase is shifted by ξ: the drive keeps the phase of the first pulse, plus ξ, so that it is $\cos(\nu t + \xi)$ and $\sin(\nu t + \xi)$, with t continuing. As in the proposition, each pulse is an exact quarter turn of $\tilde{\mathbf{r}}$, about $\mathbf{e}_x$ for the first and about $(\cos\xi, \sin\xi, 0)$ for the second (§ 2.4). (i) Show that $\mathrm{Prob}(\downarrow) = \cos^2\big((\Delta T - \xi)/2\big)$. (ii) Take $\xi = 0$ and $T = 1$ ms. How far apart in $\nu/2\pi$ are the fringes, and what is the full width at half maximum of the central one? (iii) Near resonance, $\mathrm{Prob}(\downarrow)$ changes with Δ at the rate $\mathrm{d}\,\mathrm{Prob}/\mathrm{d}\Delta$. Which ξ makes this rate at $\Delta = 0$ as large as possible in size, and what is $\mathrm{Prob}(\downarrow)$ there?

Exercise 5 (the shift of a linear drive, an estimate ⋆). In Remark 2.5, a linear drive adds to the circular one the counter-rotating term $\frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x - \sin(\nu t)\,\sigma_y\big)$. (i) Show that in the frame $\psi = e^{+\mathrm{i}\nu t\sigma_z/2}\chi$, which turns with the counter-rotating field, that term becomes $\frac{\hbar\Omega}{2}\sigma_x$ and the static term becomes $\frac{\hbar}{2}(\nu_0 + \nu)\sigma_z$. (ii) Ignore the co-rotating term in this frame. Then the spin precesses about $(\Omega, 0, D)/\sqrt{D^2 + \Omega^2}$, $D = \nu_0 + \nu$, at the rate $\sqrt{D^2 + \Omega^2}$. Show that for $\Omega \ll D$ this rate exceeds D by about $\Omega^2/(2D)$. (iii) Near resonance $D \approx 2\nu_0$. Conclude that the counter-rotating term raises the spin's own frequency, and so the resonance, by about $\Omega^2/(4\nu_0)$, and compare with a numerical solution for $\Omega/\nu_0 = 0.08$. Over one period $T = 2\pi/\nu$ of the drive, the evolution operator is computed numerically. The angle between its two eigenvalues, divided by T, is the rate at which the spin effectively turns, and the resonance is taken where this rate is smallest. This puts the resonance at $\nu/\nu_0 = 1.0016$, to four decimal places.